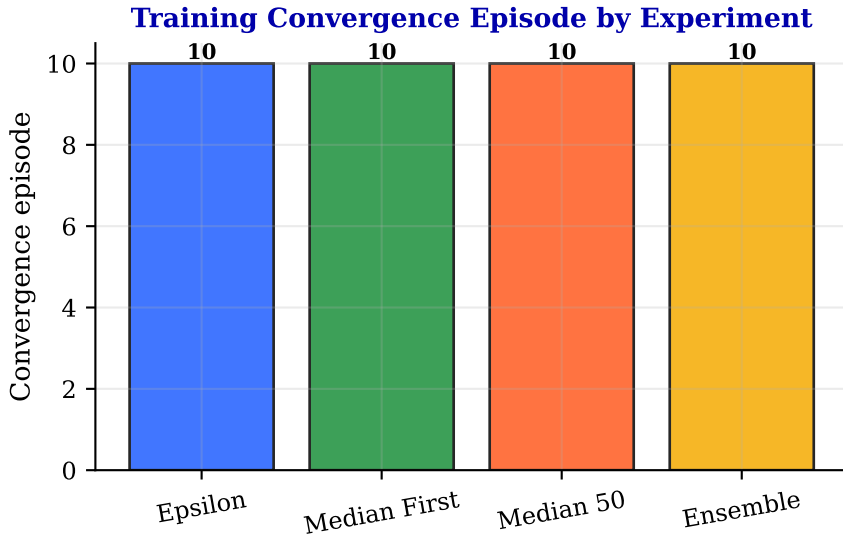
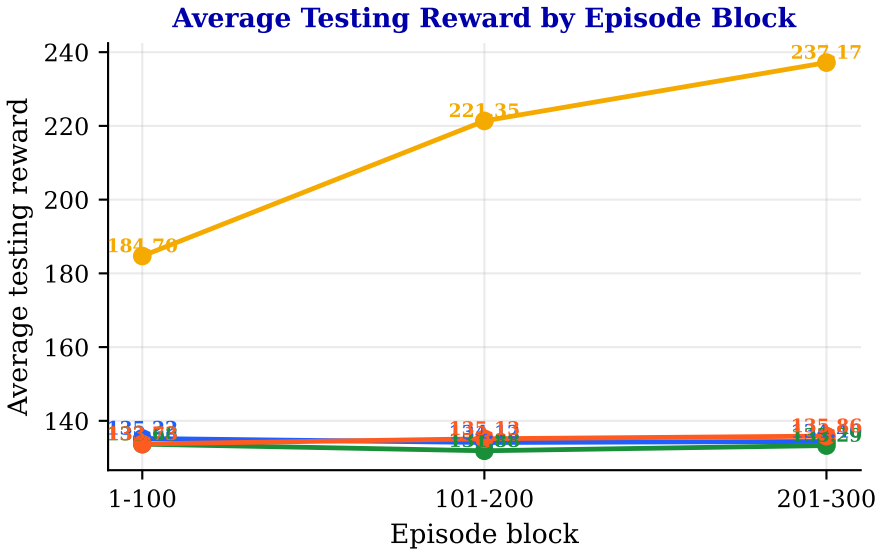
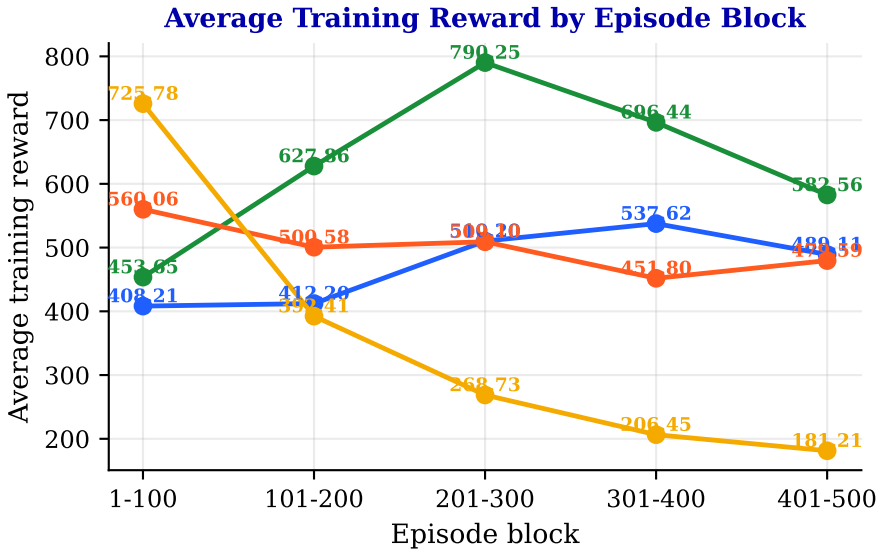


DQN in CARLA - Training & Testing Dashboard (LR multiplier = 1, Final LR = 0.0001)

All rewards are ENVIRONMENT rewards only • Training Episodes: 500 • Testing Episodes: 300 • Episode Block Size: 100 • Epsilon: 0.2



Training Reward Summary Statistics (All 500 Episodes)

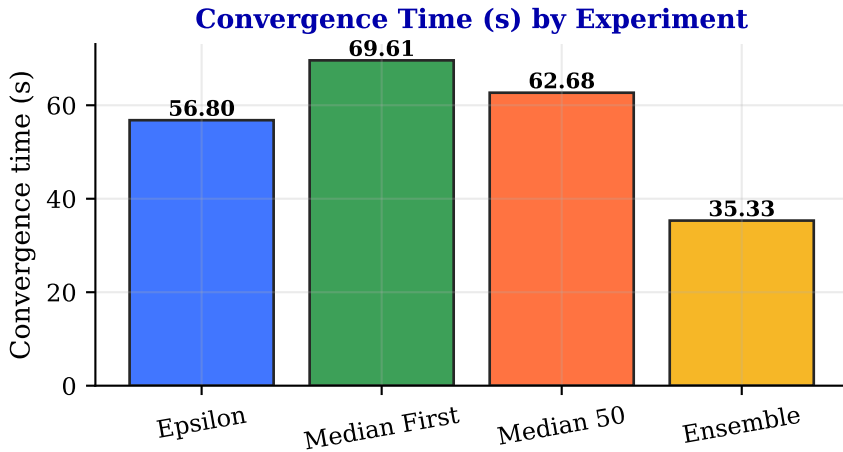
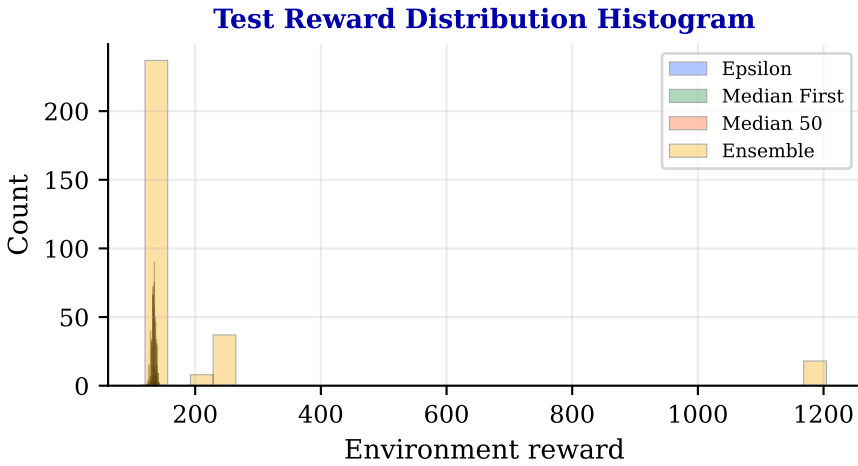
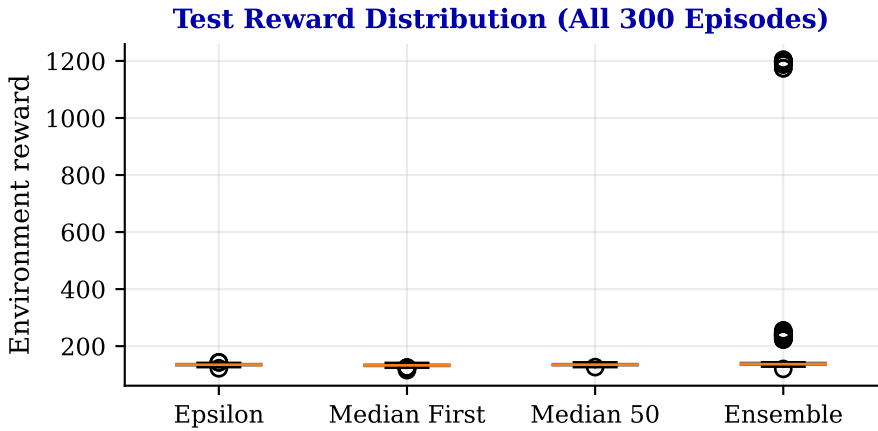
Method	Mean	Median	Std	Min	Max	Q1	Q3
Epsilon	471.47	132.76	373.33	-97.99	1110.04	126.42	846.12
Median First	630.15	738.95	416.69	115.86	1145.57	126.46	1043.79
Median 50	500.23	606.85	355.12	120.16	1127.68	126.41	819.61
Ensemble	354.92	124.32	450.97	115.98	1272.24	122.26	126.50

Testing Reward Summary Statistics (All 300 Episodes)

Method	Mean	Median	Std	Min	Max	Q1	Q3
Epsilon	134.58	134.86	3.09	122.37	143.32	132.79	136.96
Median First	132.95	132.84	3.66	115.77	141.15	130.73	134.93
Median 50	134.91	134.89	2.83	126.42	143.16	132.80	136.95
Ensemble	214.41	136.93	250.09	120.11	1204.63	134.87	141.13

Convergence Summary

Method	Conv Episode	Train Episodes	Train Time(s)	Conv Time(s)
Epsilon	10	500	2,840.07	56.80
Median First	10	500	3,480.30	69.61
Median 50	10	500	3,134.19	62.68
Ensemble	10	500	1,766.29	35.33



Key Takeaways

- Ensemble has the highest average test reward (214.41).
- Ensemble has the fastest convergence time (35.33s).
- Median 50 has the lowest test reward std (2.83).
- Test networks are frozen; no updates occur during testing.

Best Overall

ENSEMBLE

Average Test Reward: 214.41
Convergence Time: 35.33s

Ranking uses reward rank + convergence-time rank.

Experiment Settings

Environment : CARLA 0.9.14 (Real)
Algorithm : DQN
Network : RND + Count + Target + Replay + Adam
NoisyNet : Removed
Learning Rate Mult. : 1
Final Learning Rate : 0.0001
Epsilon : 0.2
Training Episodes : 500
Testing Episodes : 300
Episode Block Size : 100